

1. Objectives

- Learn to create a smart system that detects nearby objects and responds with visual feedback.
- Learn to understand the principles of ultrasonic sensors, becoming familiar with the Raspberry Pi and its components.
- Learn to measure distances accurately and understand how it communicates with the Raspberry Pi.
- Make the LED light up when an object comes within 5 cm of the sensor.

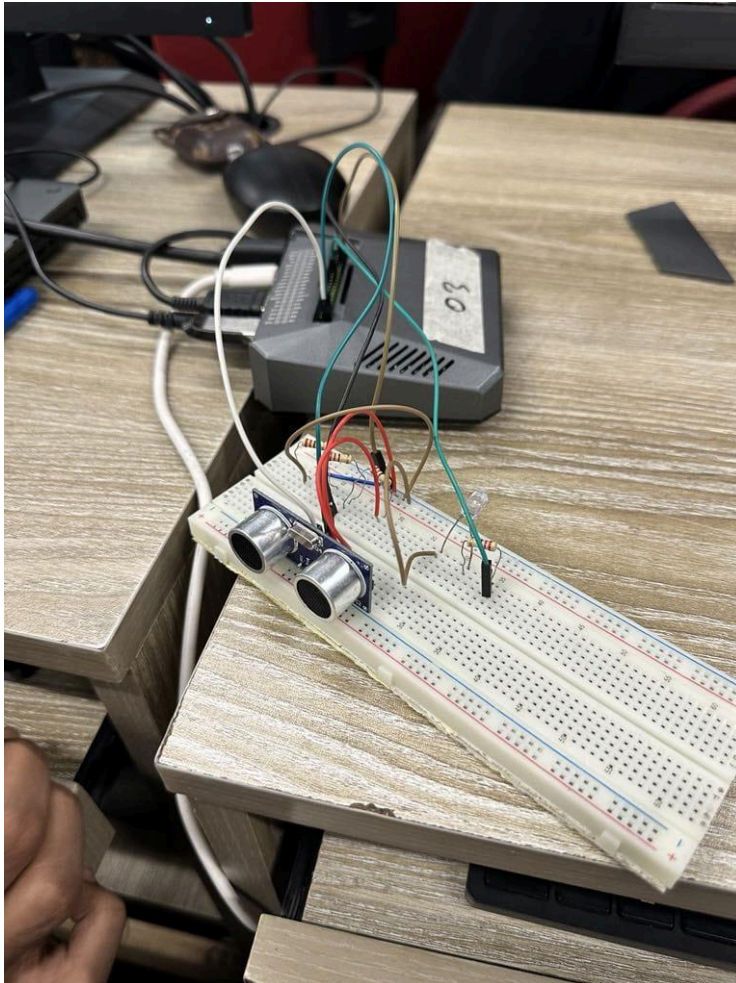
Learn how to protect the Raspberry Pi from high sensor signals by setting up a voltage divider.

- The goal is to be able to measure distance using the ultrasonic sensor.

2. Equipments: For measuring distance using an ultrasonic sensor on the Raspberry Pi 4, we need the following electronic components:

- Raspberry pi 4.
- Ultrasonic Sensor (HC-SR04).
- Breadboard, Connecting wire (Male to Female, Male to male, Female to Female).
- 1k and 1.5k resistors.
- MicroSD Card, USB Cable.

3. Experimental Setup: (Picture + Explanation)



```
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task1.py X task2.py X
1 import RPi.GPIO as GPIO
2 import time
3 from gpiozero import LED
4 GPIO.setmode(GPIO.BCM)
5 TRIG = 23 # GPIO pin used to send square wave
6 ECHO = 24 # GPIO pin used to receive square wave
7 led = LED(17)
8
9 GPIO.setmode(GPIO.BCM)
10 GPIO.setmode(GPIO.BCM)
11 GPIO.setmode(GPIO.BCM)
12
13 def distance():
14     GPIO.output(TRIG, False)
15     time.sleep(0.1)
16     GPIO.output(TRIG, True)
17     time.sleep(0.00001)
18     GPIO.output(TRIG, False)
19     pulse_start = time.time()
20
21     while GPIO.input(ECHO) == 0:
22         pulse_start = time.time()
23     while GPIO.input(ECHO) == 1:
24         pulse_end = time.time()
25         pulse_duration = pulse_end - pulse_start
26         distance = pulse_duration * 17150
27         distance = round(distance, 2)
28         if distance == 0:
29             led.on()
30             print("led on")
31         else:
32             led.off()
33
34     return distance
35
36 while True:
37     print(distance())
38     GPIO.cleanup()
39
40 Shell
41 led on
42 3.00
43 led on
44 4.10
45 led on
46 2.90
47 led on
48 2.71
```

We placed the Ultrasonic Sensor (HC-SR04) and resistors in the breadboard. The Echo pin of the Ultrasonic Sensor is connected to one side of the two resistors connected in series. And the other side of the resistors is connected to the GPIO 20 pin, and the other three resistors are also connected in series. These three resistors' other side is connected to the GND pin of the ultrasonic sensor. whereas the GND pin is also connected to the 06 no. Pin, labeled with Ground, of the raspberry pi 4 by a male-to-female wire. The Trigger pin of the Ultrasonic Sensor is connected to the GPIO 21 pin, and the Vcc pin is connected to the 04 no. Pin, labeled with 5V DC Power, of the raspberry pi 4 by a male-to-female wire. After connecting everything and running the following code, the Ultrasonic Sensor (HC-SR04) showed the measured distance.

4. Code: (If Applicable)

```
# Enter Code Here

import RPi.GPIO as GPIO
import time

from gpiozero import LED
GPIO.setmode(GPIO.BCM)
TRIG = 21 #GPIO21
ECHO = 20 #GPIO20
Led = LED(17)

GPIO.setup(TRIG, GPIO.OUT)
GPIO.setup(ECH, GPIO.IN)

def distance():
    GPIO.output(TRIG, False)
    time.sleep(0.5)
    GPIO.output(TRIG, True)
    time.sleep(0.00001)
    GPIO.output(TRIG, False)
    Pulse_start = time.time()

    while GPIO.input(ECHO) == 0:
        pulse_start = time.time()
    while GPIO.input(ECHO) == 1:
        pulse_end = time.time()
    pulse_duration = pulse_end - pulse_start

    distance = pulse_duration * 17150
    distance = round(distance,2)
```

```
if distance <= 5:
    led.on()
    print("led on")
else:
    led.off()
return distance

while True:
    print(distance())

GPIO.cleanup()
```

5. Results (Output of the experiment):

According to our setup, the possible result for measuring distance using an ultrasonic sensor is displayed in the Python terminal in Raspberry Pi 4. When an object was in a constant place, we got the constant distance value after pressing the run button of the terminal. Again, when we increased the object's distance from the ultrasonic sensor, we got a higher distance value than the previous one. Furthermore, when we decreased the object's distance from the sensor, we got a lower distance value than the previous.

6. Discussions/Answers:

(May contain conclusions or learnings from the result along with problems faced and solving methodology.)

This experimental task was to measure the distance of an object by using an ultrasonic sensor. This task taught us how to simultaneously use a raspberry pi, ultrasonic sensor, and breadboard and the functionality of different raspberry pi and ultrasonic sensor pins. Moreover, we get to familiarize ourselves with the GPIO and time library. Fortunately, we connected all the task components without a problem and got our desired result.